The quilt of unmet need: patch area is how common the disease is, colour is how much of its chromatin anyone has read

Each patch is one of the 45 named entities, grouped into three regions by age class and sized within them by US incidence. The same tiling is coloured two ways. Colour runs loud-to-quiet as coverage improves, so the entities with the least data are the ones that reach the eye first; a patch that quietens from A to B had only looked well-covered because the disease is rare.

samples (panel A) — least data on the left

1 4,327

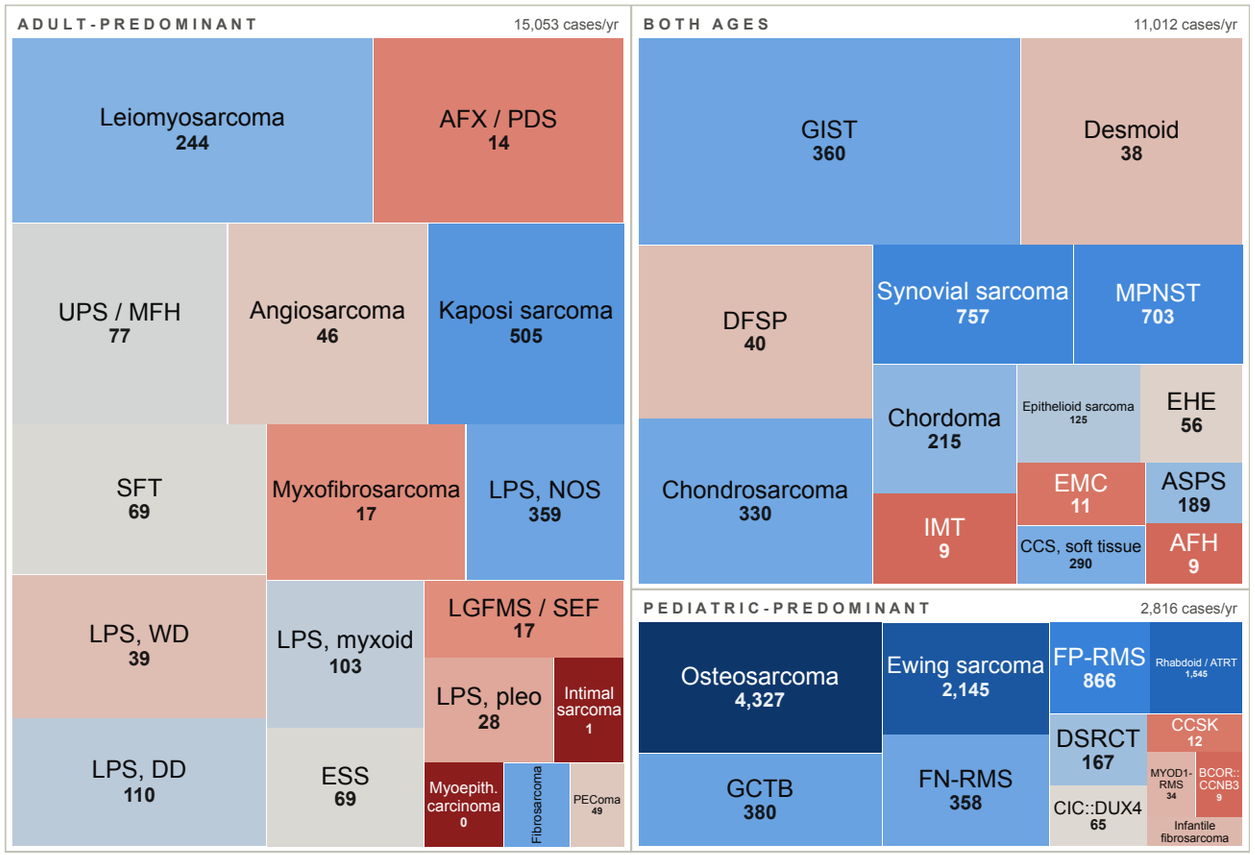
per case (panel B) — least data on the left

0.01 16.3

cross-hatched: no epigenomic data of any kind

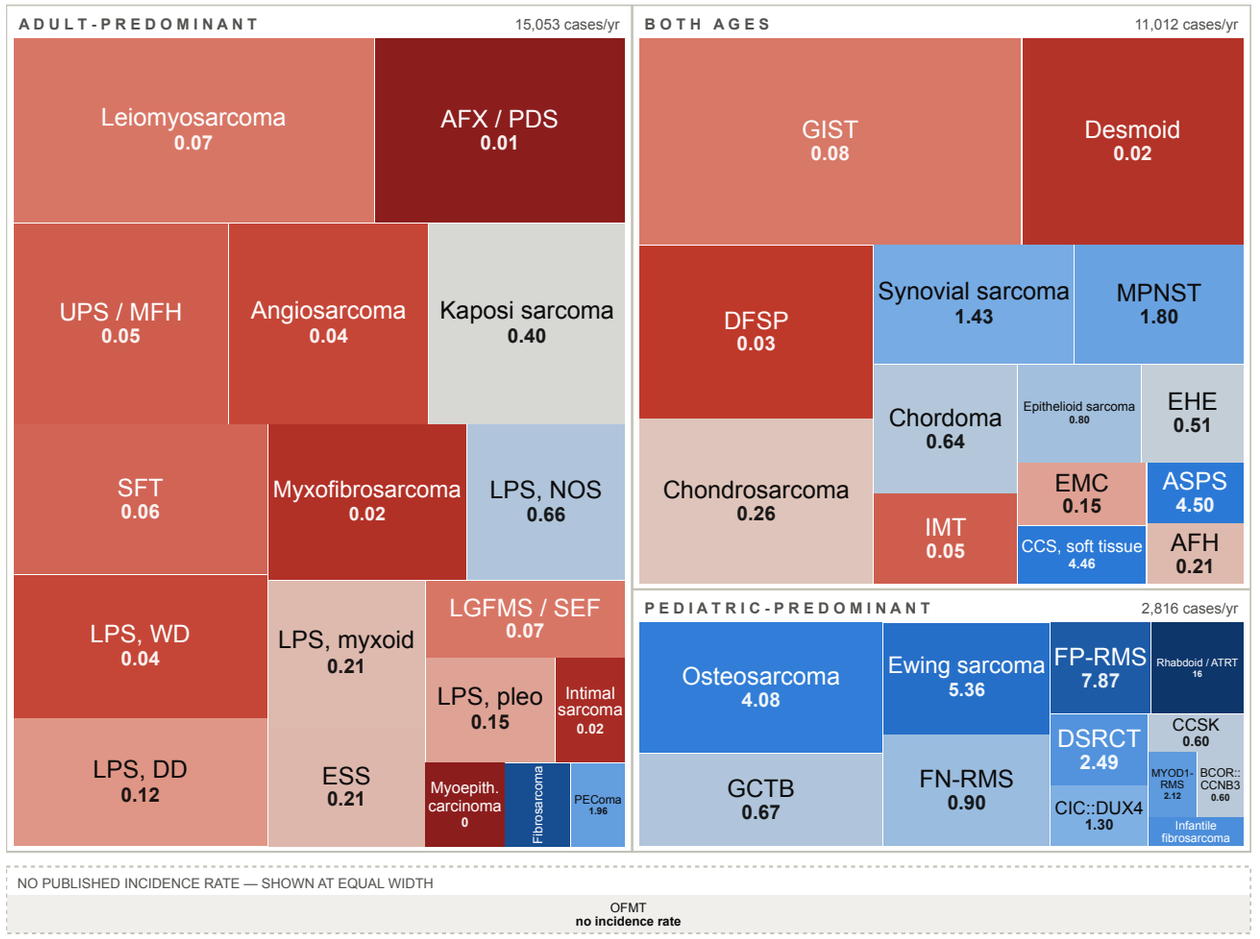
A Coloured by epigenomic samples

absolute count, log scale · 1 to 4,327 · loud = least data



B Coloured by samples per new case per year

coverage against need, log scale · 0.01 to 16.3 · loud = least data



Patch area is proportional to (US cases per year) raised to the power 0.55, which compresses a 359-fold spread in incidence to roughly 25-fold, so that the rarest entities remain legible. Area still ranks burden correctly, but differences in area understate differences in incidence; the case count is printed on every patch that has room for it. Colour is log-scaled in both panels and runs loud to quiet as coverage improves; an entity with no epigenomic data of any kind takes the loud end and is cross-hatched, because none is a different statement from few. Epigenomic means every assay in the atlas, DNA methylation included. Incidence is US cases per year over the 45 named entities; 1 entity has no population rate published anywhere and is shown at equal width in the strip beneath, excluded from every area and from panel B.